

● EASY

● ALGEBRA

Linear functions

07

10 questions · ~15 min

Q1

ALGEBRA

$$f(x) = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

- A. The tree will be measured each year for 4 years.
- B. The tree is estimated to grow to a maximum height of 4 feet.
- C. The estimated height of the tree increased by 4 feet each year.
- D. The estimated height of the tree was 4 feet when it was first measured.

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of \$ 58?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 2 and has a y -intercept at 0, - 5. Which equation defines f ?

A. $f(x) = \frac{1}{2}x - 5$

B. $f(x) = -\frac{1}{2}x - 5$

C. $f(x) = -2x - 5$

D. $f(x) = 2x - 5$

Sean rents a tent at a cost of \$ 11 per day plus a onetime insurance fee of \$ 10. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

A. $c = 11d + 10$

B. $c = 10d + 11$

C. $c = 11d + 10$

D. $c = 10d + 11$

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

- A. Tavon will keep the vine plant for 9 months.
- B. The vine plant is expected to grow 9 inches each month.
- C. The vine plant is expected to grow to a maximum length of 9 inches.
- D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

The function g is defined by $g(x) = -x + 8$. What is the value of $g(0)$?

A. -8

B. 0

C. 4

D. 8

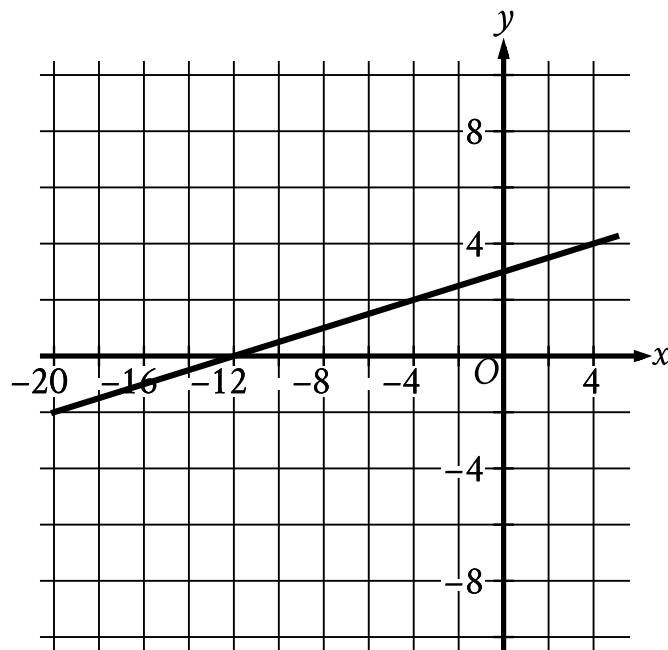
For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

A. $f(x) = 3x$

B. $f(x) = 3x - 8$

C. $f(x) = 3x + 5$

D. $f(x) = 3x + 11$



- The line slants gradually up from left to right.
- The line passes through the following points:
 - (negative 12 comma 0)
 - (negative 4 comma 2)

The graph of the linear function f is shown, where $y = fx$. What is the x -intercept of the graph of f ?

- A. -12,0
- B. 0,0
- C. $\frac{1}{4},0$
- D. 12,0

$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A. 40
- B. 43
- C. 45
- D. 55